INCH-POUND

MIL-I-24728/1A
23 July 1992
SUPERSEDING
MIL-I-24728/1(NAVY)
31 August 1989

MILITARY SPECIFICATION SHEET

INTERCONNECTION BOX, FIBER OPTIC, SUBMERSIBLE, 254 X 330 mm

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24728.

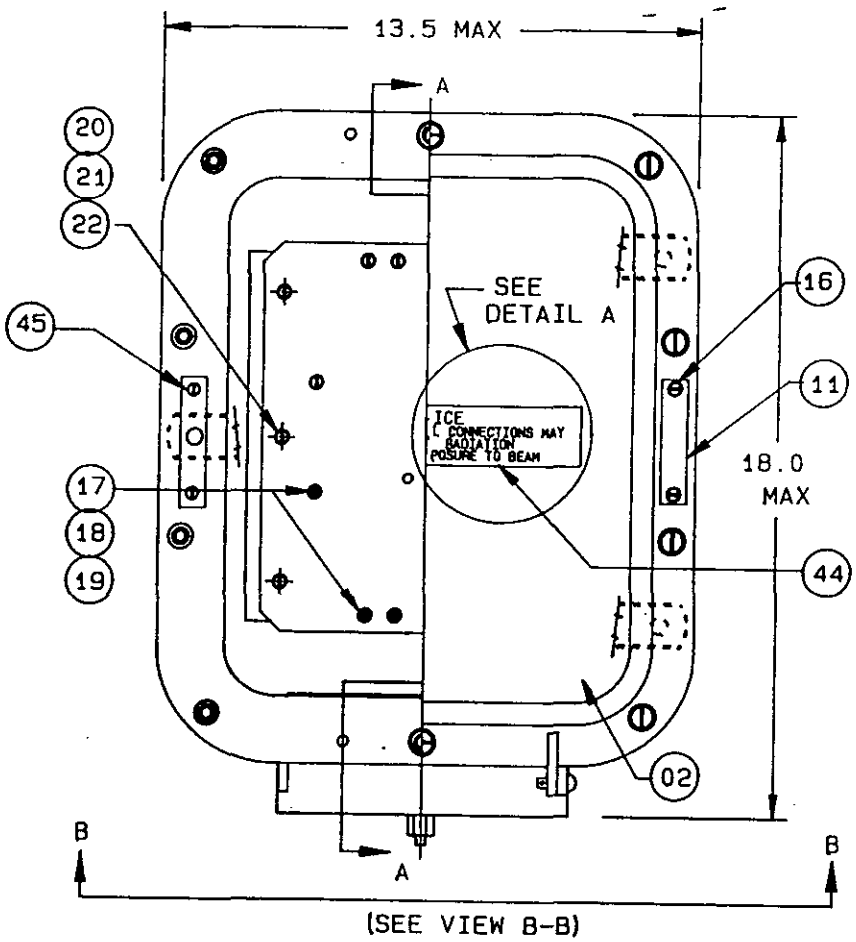


FIGURE 1. Enclosure assembly, M24728/1-001.

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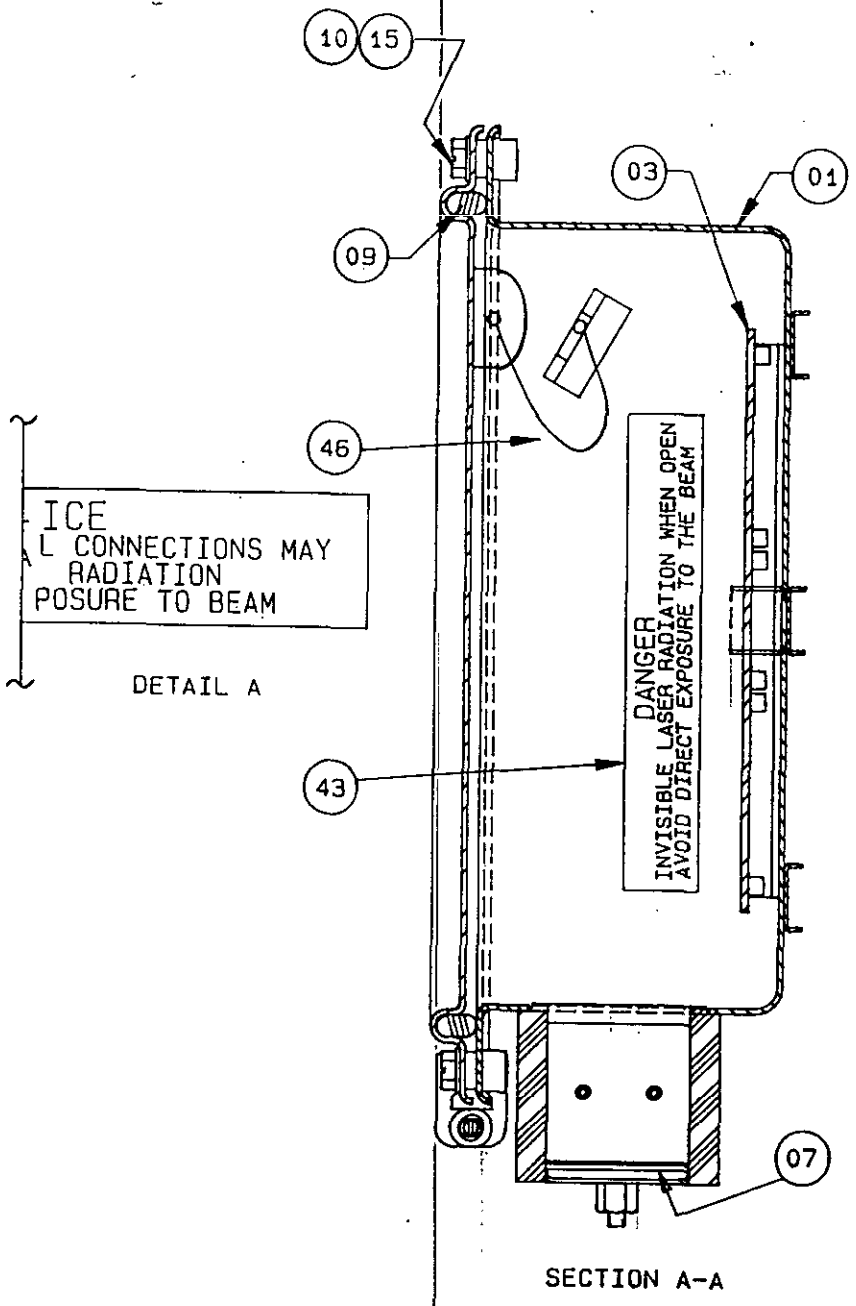
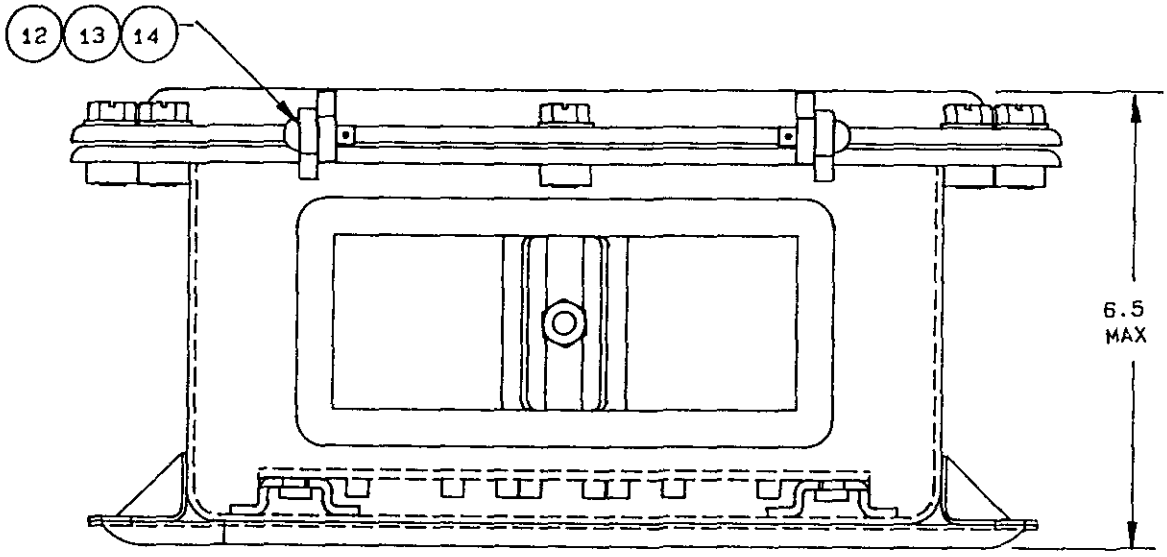


FIGURE 1. Enclosure assembly, M24728/1-001 - Continued.

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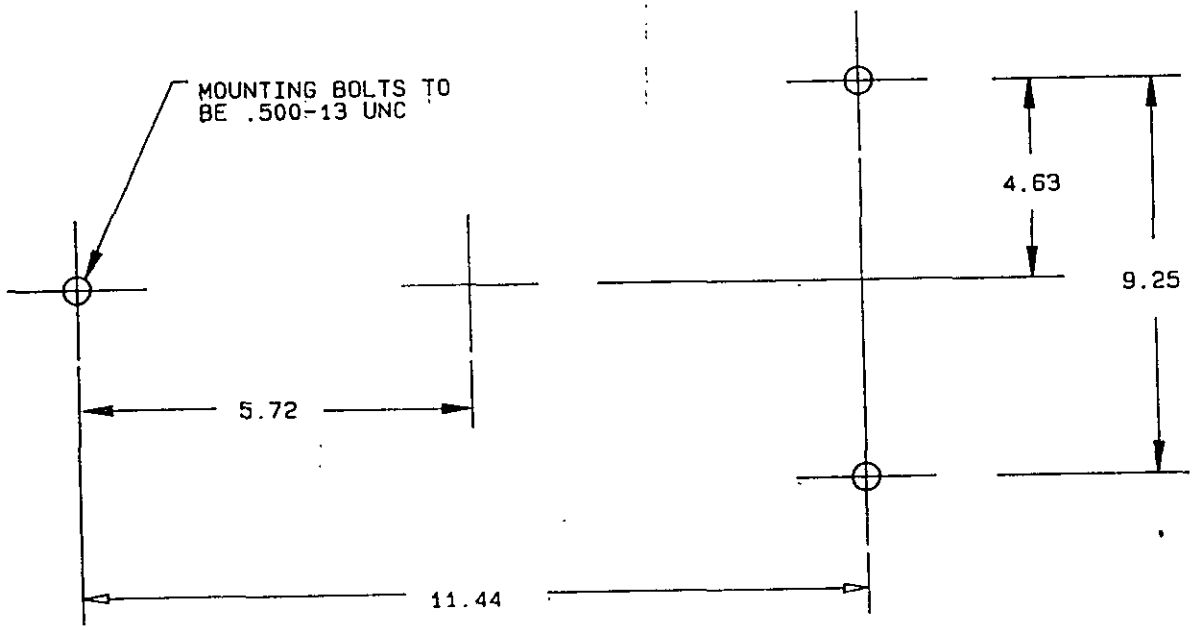


Inches	mm
6.5	165
13.5	343
18.0	457

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals.

FIGURE 1. Enclosure assembly, M24728/1-001 - Continued.

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Inches	mm
.500	12.70
4.63	117.6
5.72	132.6
9.25	235.0
11.44	290.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals and ±.005 (0.13) for 3 place decimals.

FIGURE 2. Mounting dimensions.

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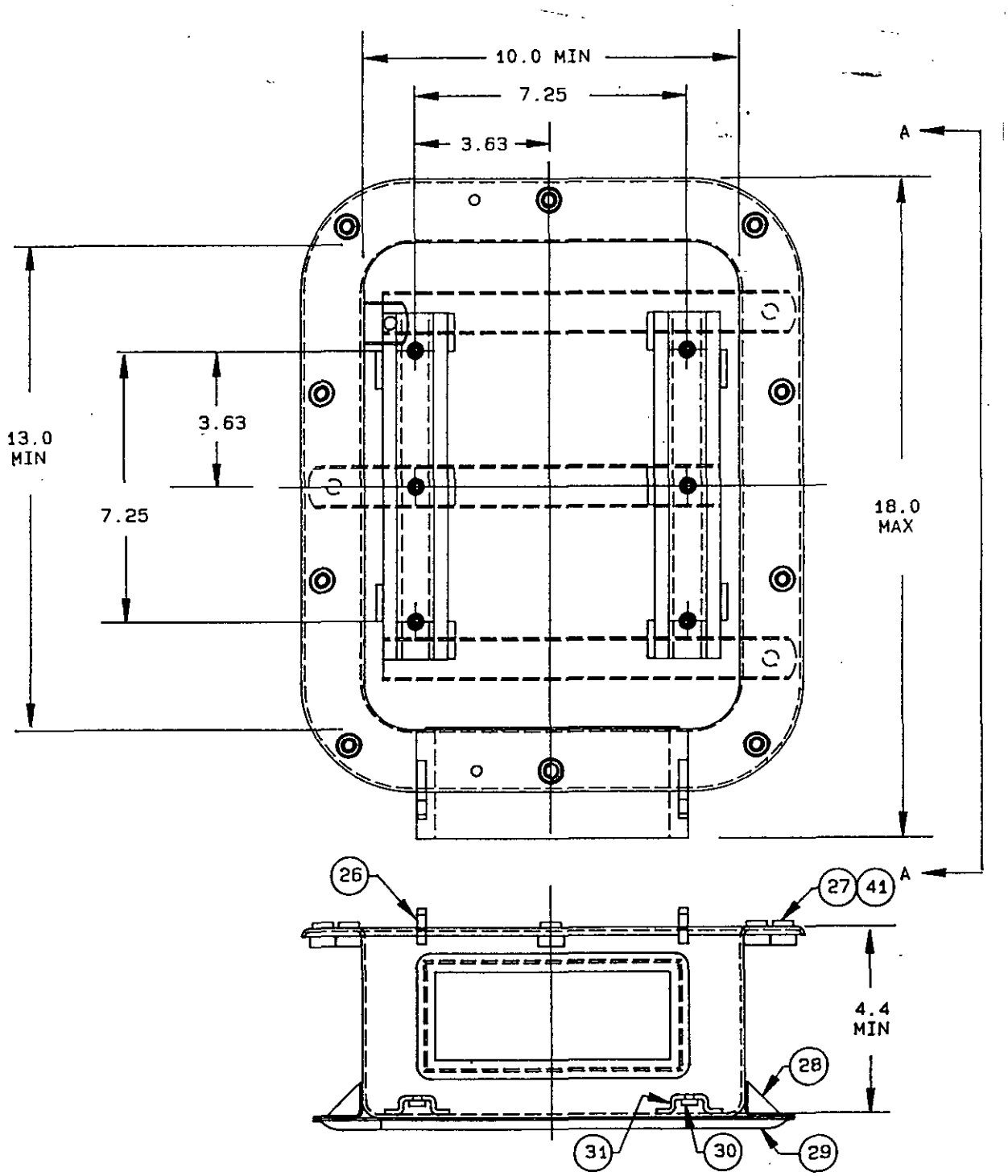
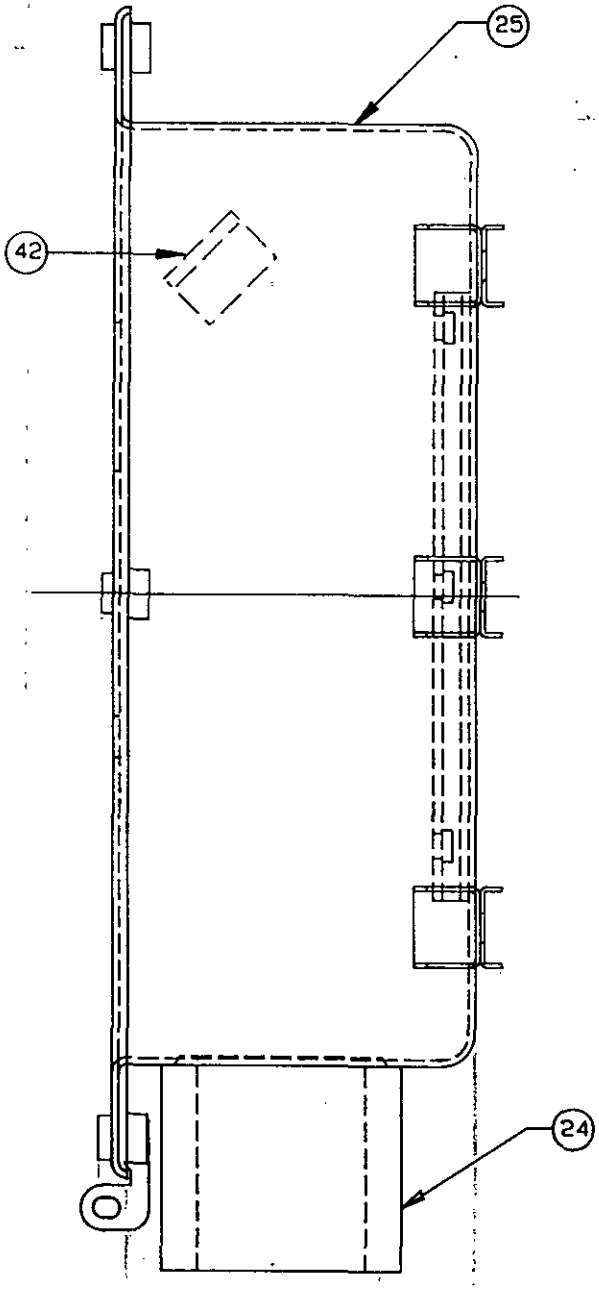


FIGURE 3. Box assembly.

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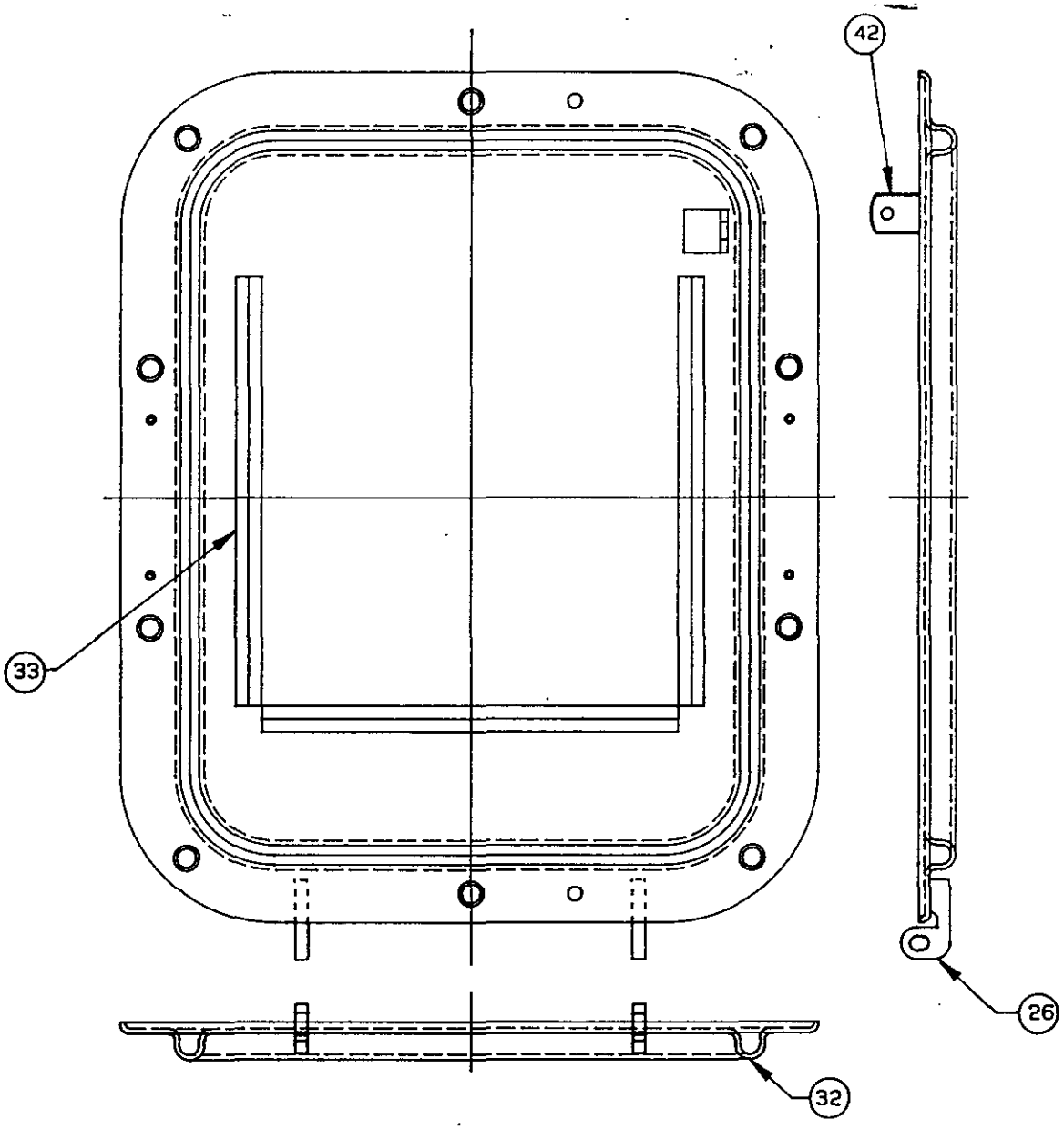
VIEW A-A

Inches	mm
3.63	92.2
4.4	112.0
10.0	254.0
13.0	330.2
7.25	184.1
18.0	457.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals.

FIGURE 3. Box assembly - Continued.

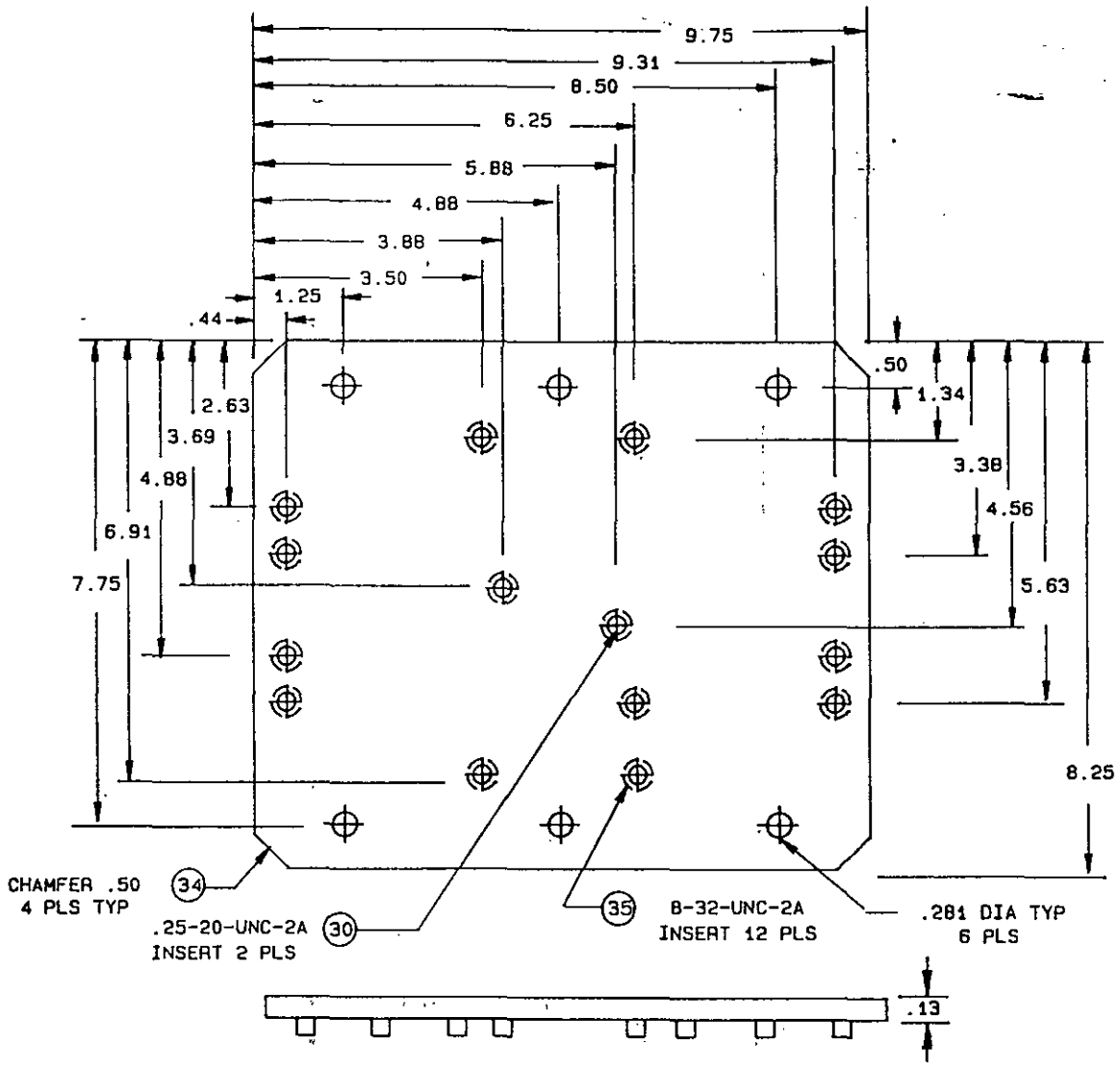
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals and ±.005 (0.13) for 3 place decimals.

FIGURE 4. Cover assembly.

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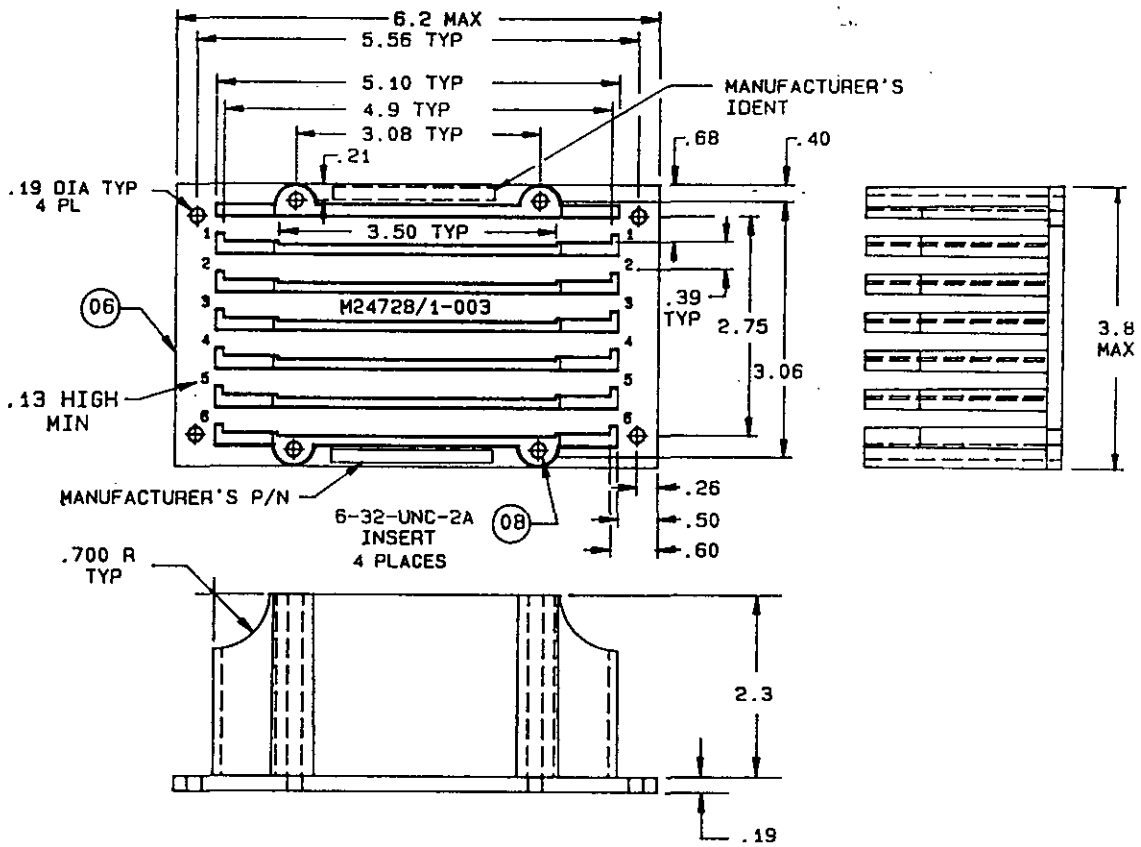
Inches	mm	Inches	mm	Inches	mm
.13	3.3	3.50	88.9	6.91	175.4
.28	7.1	3.69	96.7	7.75	196.9
.44	11.1	3.88	98.4	8.25	209.6
.50	12.7	4.56	115.9	9.31	236.6
1.25	31.8	4.88	123.8	9.75	247.6
1.34	34.0	5.63	142.9		
2.53	64.1	5.88	149.2		
2.63	66.8	6.25	158.8		
3.38	85.7				

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals and ±.005 (0.13) for 3 place decimals.

FIGURE 5. Mounting plate assembly.

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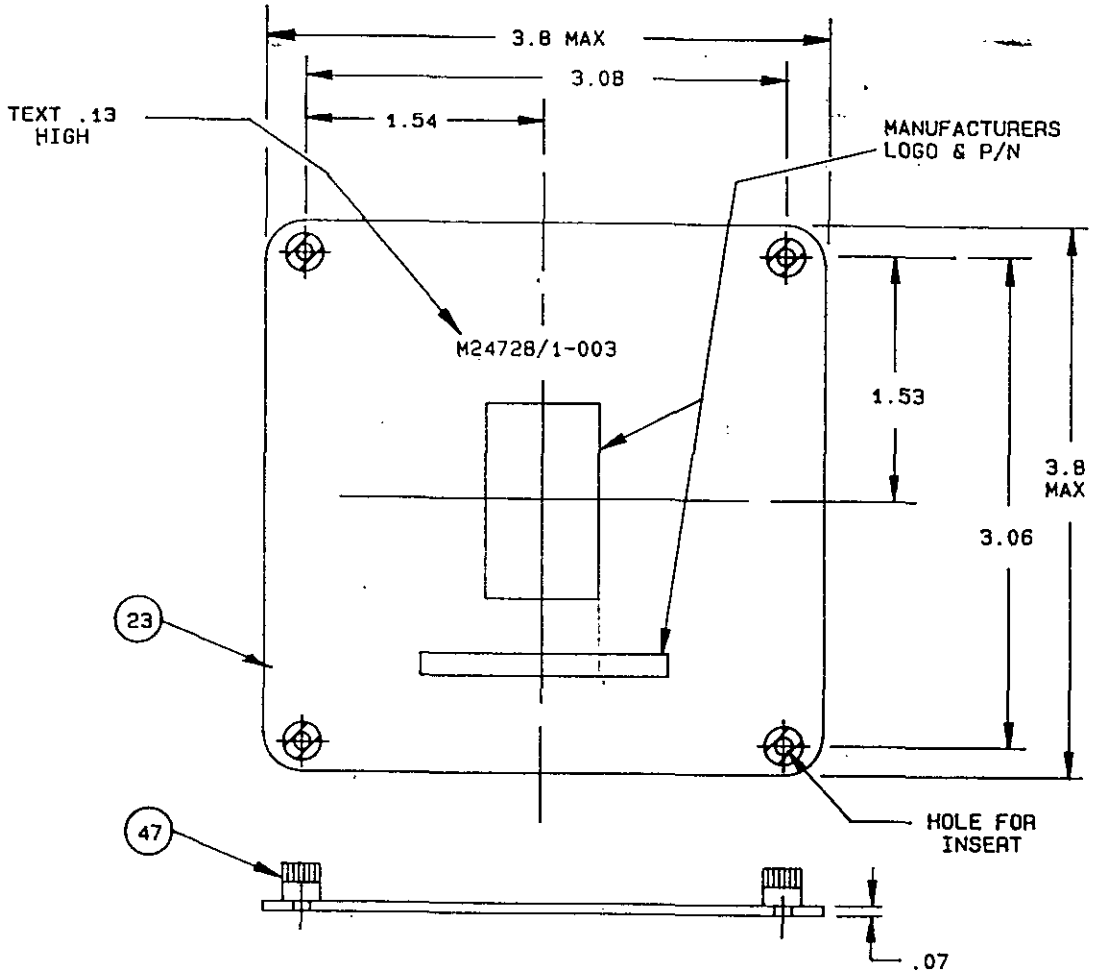


Inches	mm	Inches	mm	Inches	mm
.13	3.3	.60	15.2	3.13	79.5
.19	4.8	.68	17.2	3.50	88.9
.21	5.3	.700	17.78	3.8	97.0
.26	6.7	2.3	58.4	4.9	124.5
.39	9.9	2.75	69.9	5.10	129.5
.40	10.2	3.06	77.7	5.56	141.3
.50	12.7	3.08	78.2	6.2	158.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals and ±.005 (0.13) for 3 place decimals.

FIGURE 6. Splice tray holder.

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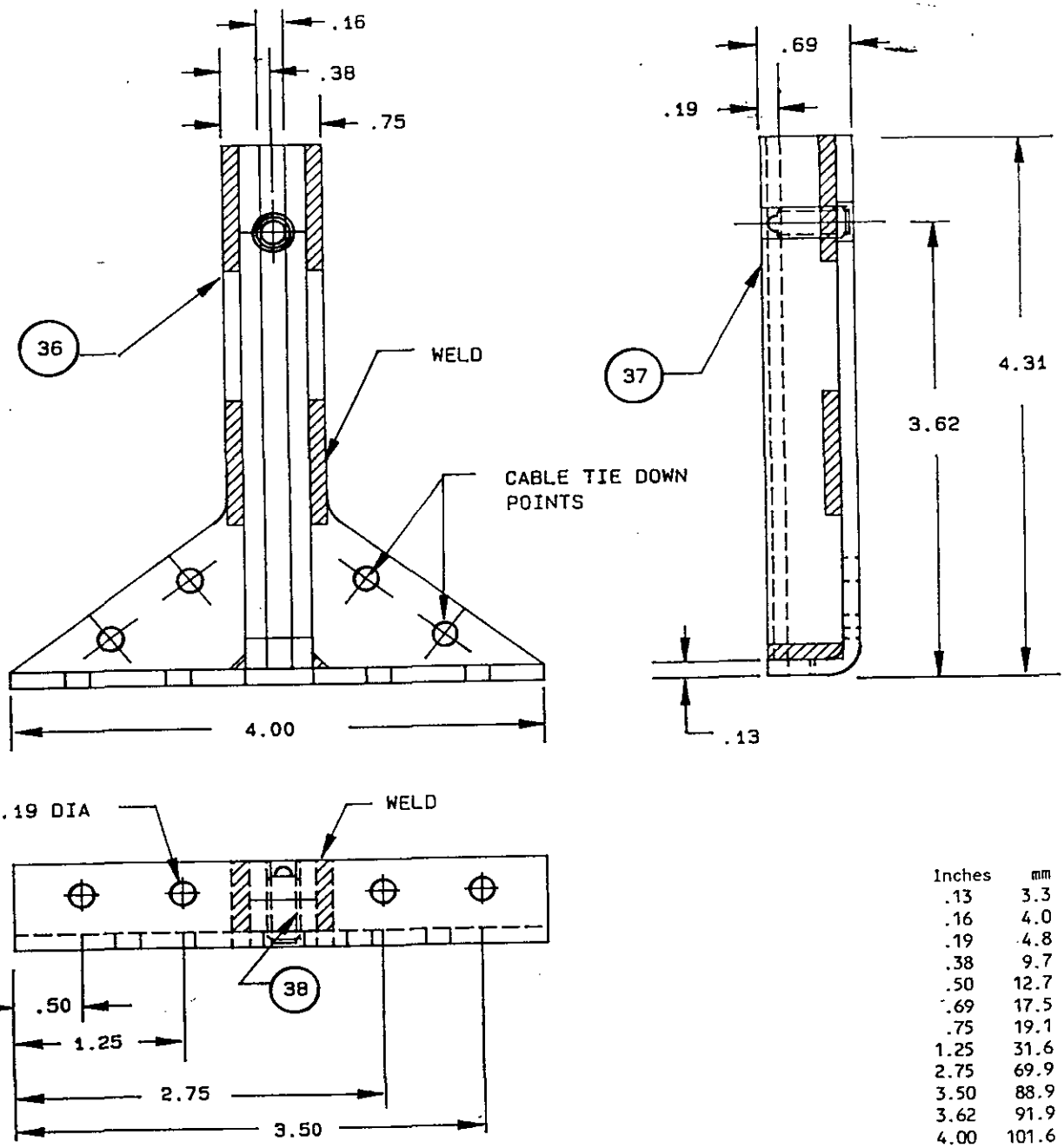


Inches	mm
.07	1.7
.13	3.3
1.53	38.8
1.54	39.1
3.06	77.7
3.08	78.2
3.8	97.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals.

FIGURE 7. Splice tray cover.

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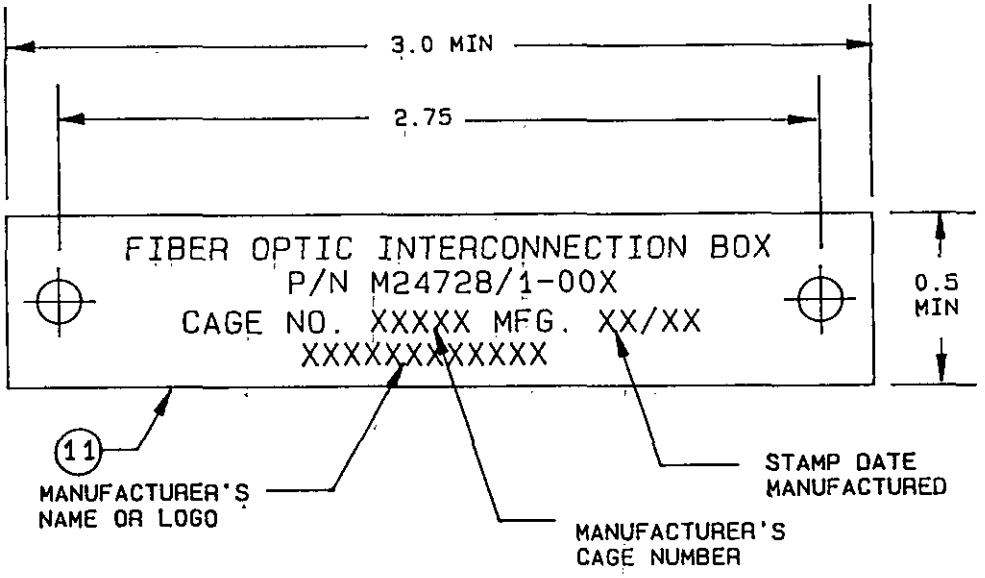


Inches	mm
.13	3.3
.16	4.0
.19	4.8
.38	9.7
.50	12.7
.69	17.5
.75	19.1
1.25	31.6
2.75	69.9
3.50	88.9
3.62	91.9
4.00	101.6
4.31	109.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals.

FIGURE 8. Slide channel assembly.

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Inches	mm
0.5	13
2.75	69.8
3.0	76.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.13) for 2 place decimals.

FIGURE 9. Identification label.

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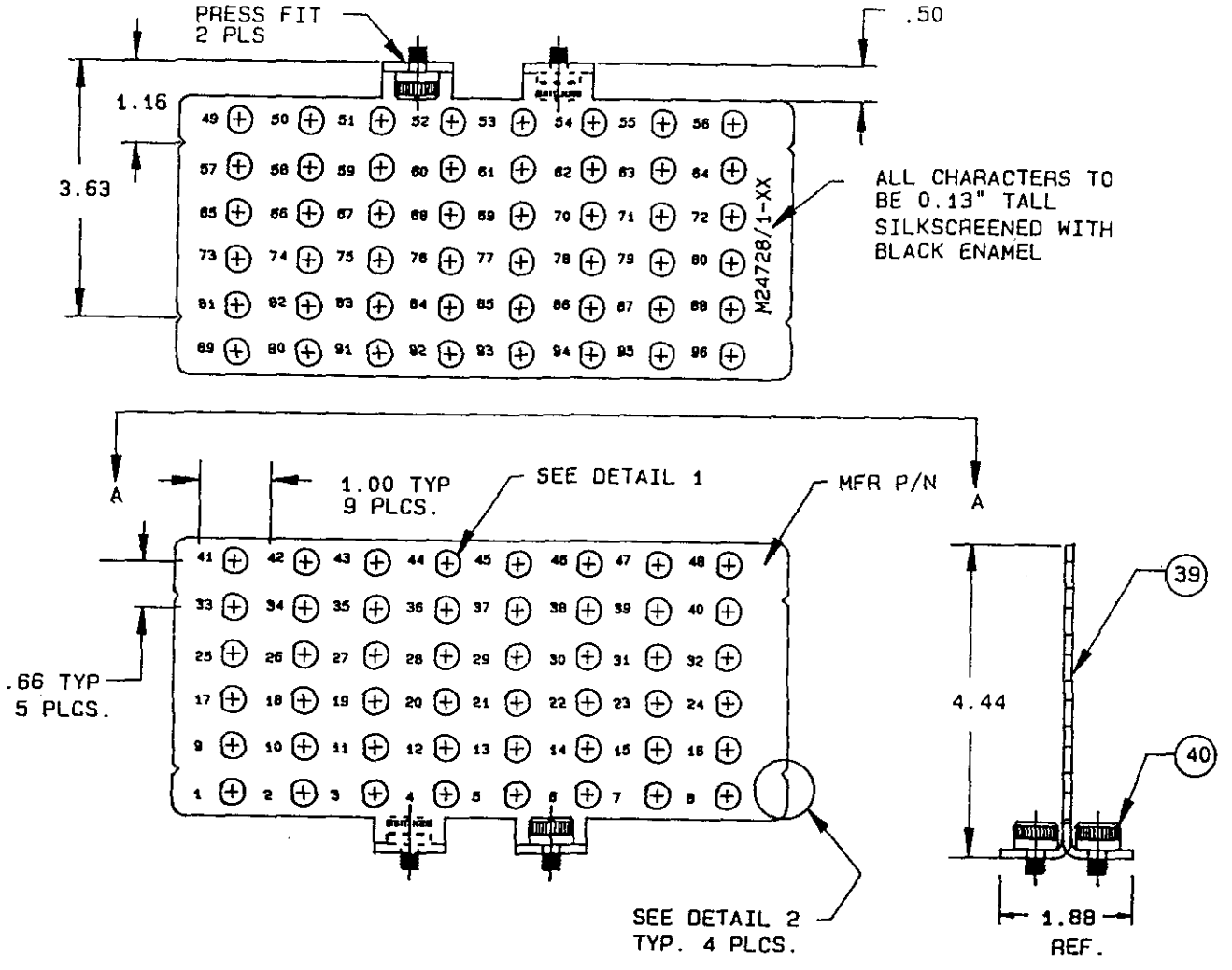
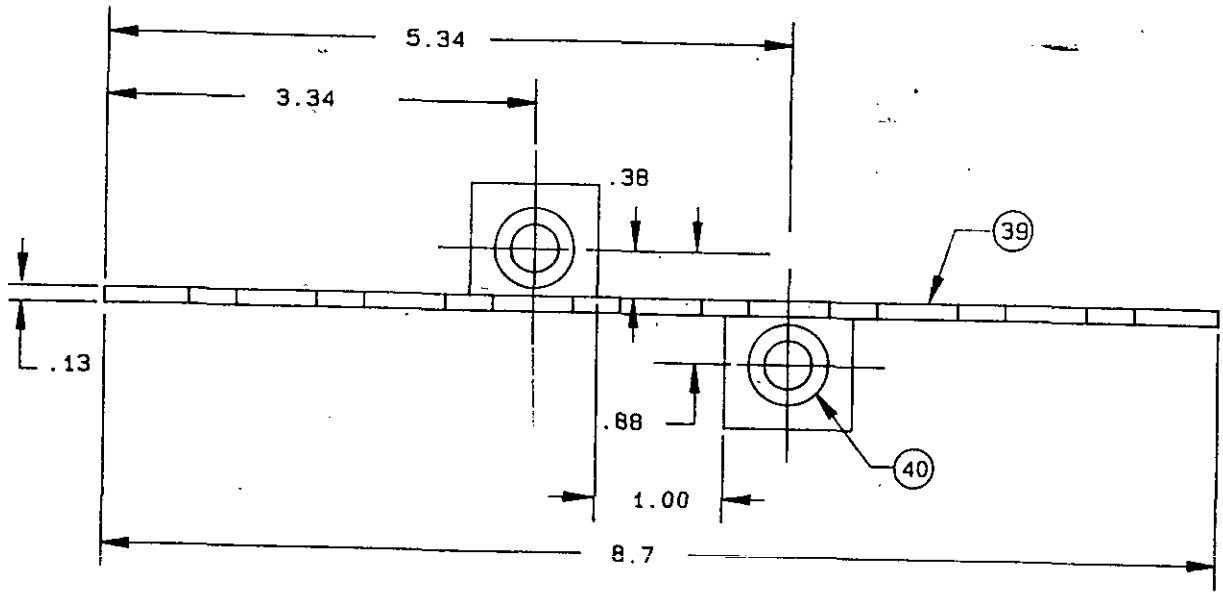
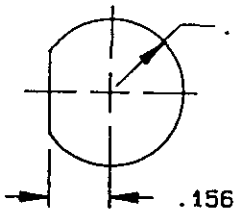


FIGURE 10. Connector optical patch panel assembly.

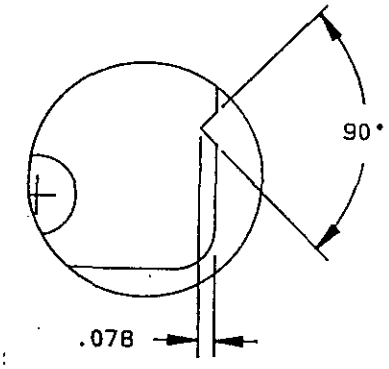
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VIEW A-A



DETAIL 1



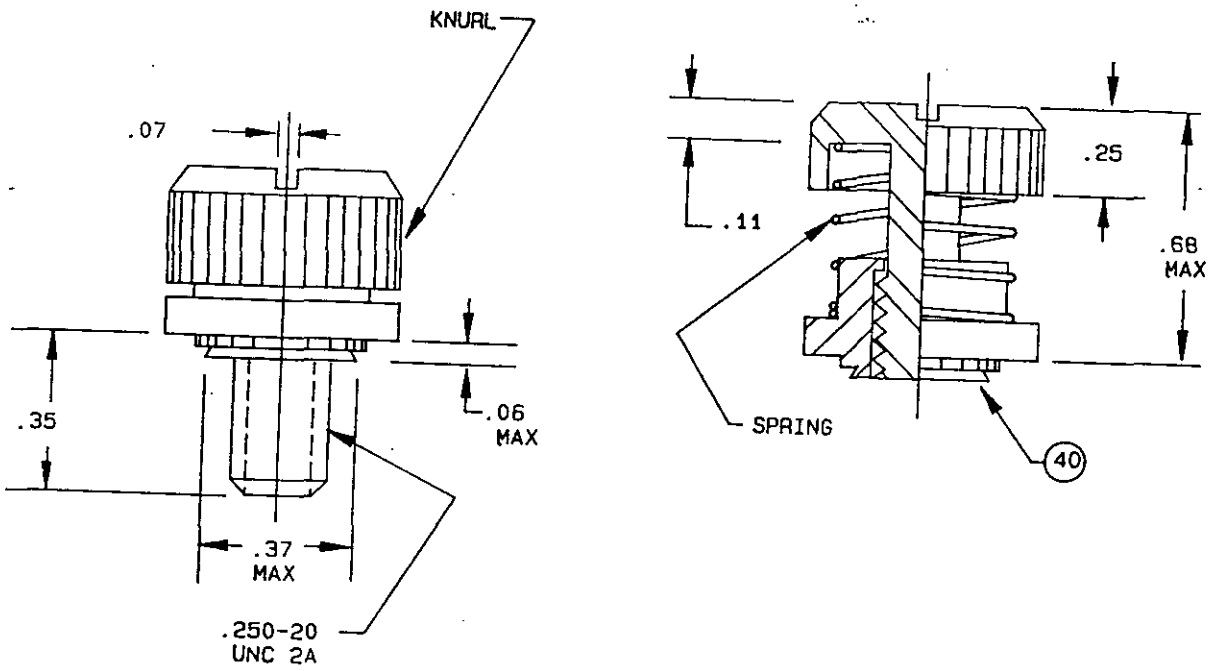
DETAIL 2

Inches	mm
.13	3.3
.078	1.98
.156	3.96
.187	4.75
.38	9.65
.50	12.7
.66	16.8
.88	22.4
1.00	25.4
1.16	29.5
1.87	47.5
1.88	47.8
3.34	84.8
3.63	92.2
4.44	112.8
5.34	121.6
8.69	220.7
8.7	221

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals and ±.005 (0.13) for 3 place decimals.
4. Tolerance for angles is ±2°.

FIGURE 10. Connector optical patch panel assembly - Continued.

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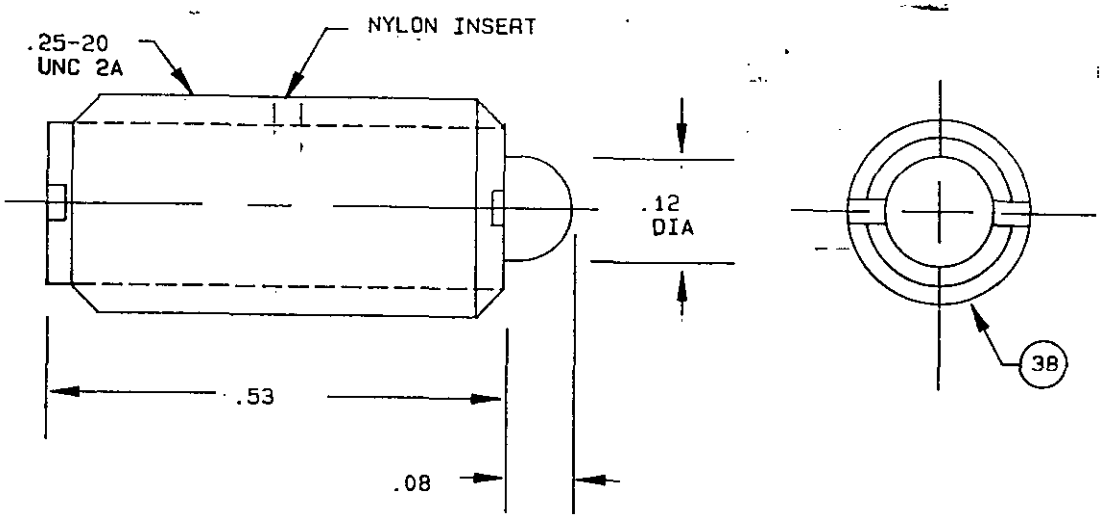


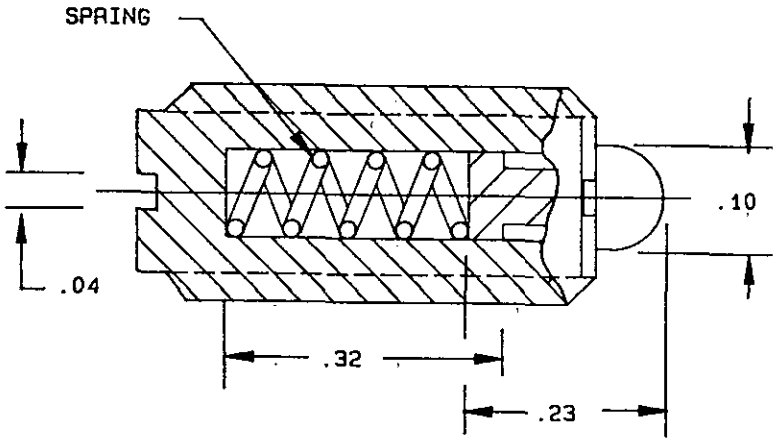
Inches	mm
.06	1.5
.07	1.8
.11	2.8
.25	6.3
.35	8.8
.37	9.4
.68	17.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals.

FIGURE 11. Captive screw.

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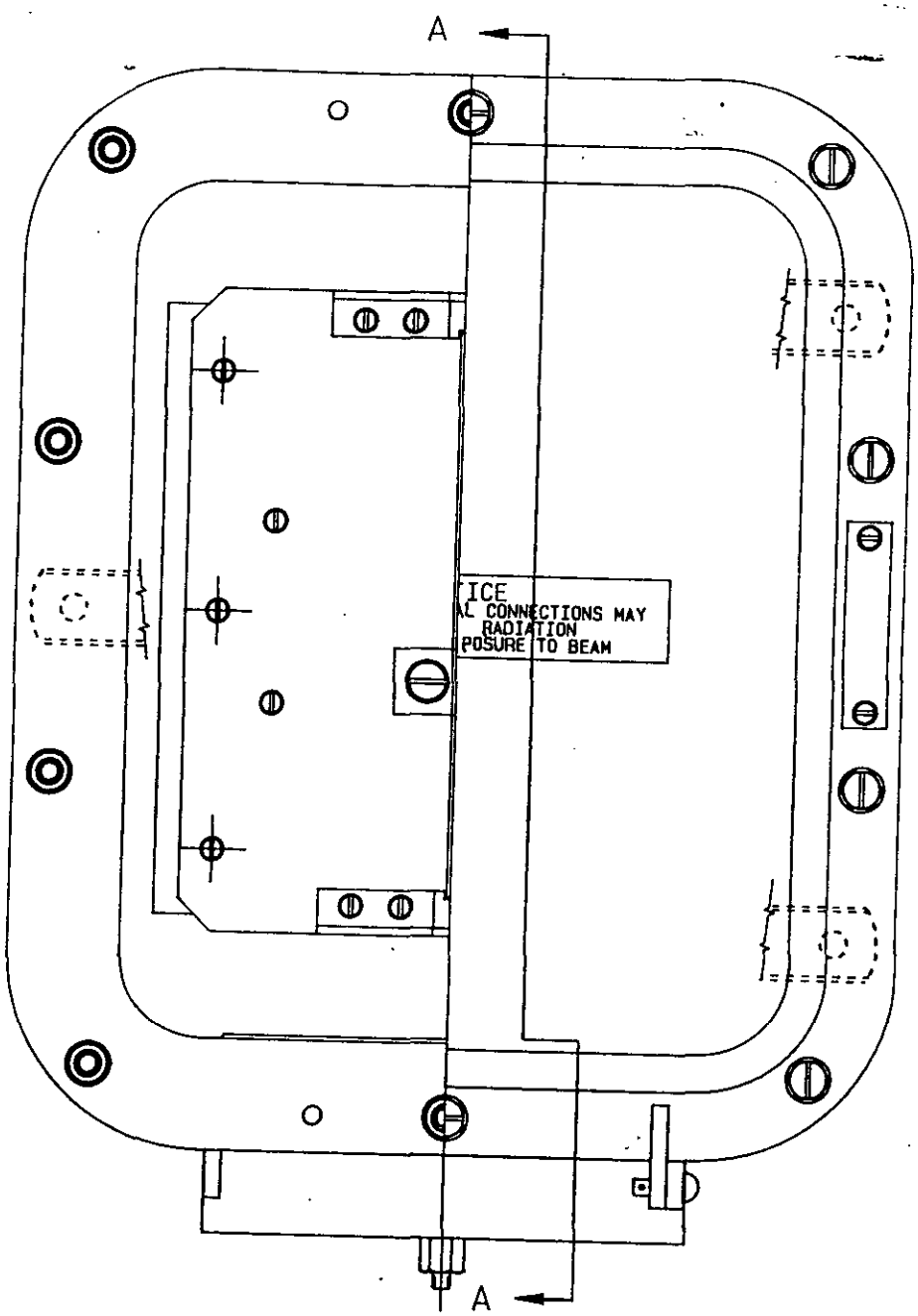


Inches	mm
.04	1.0
.08	2.0
.10	2.5
.12	3.0
.23	5.8
.25	6.35
.32	8.1
.53	13.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals.

FIGURE 12. Spring plunger.

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ABOVE COVER SHOWN PARTIALLY REMOVED

FIGURE 13. Patch panel kit M24728/1-002 installed in an enclosure assembly, M24728/1-001.

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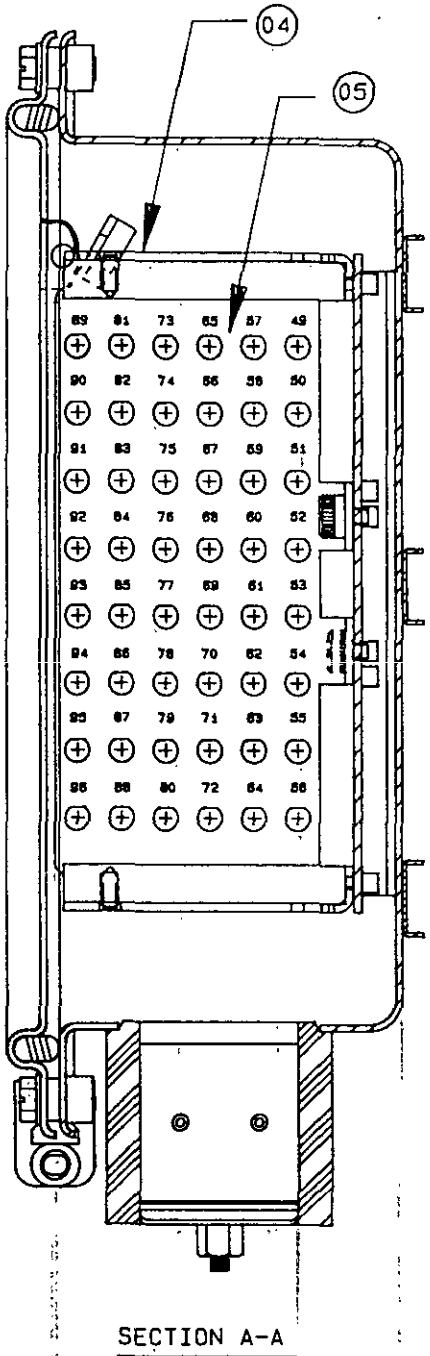
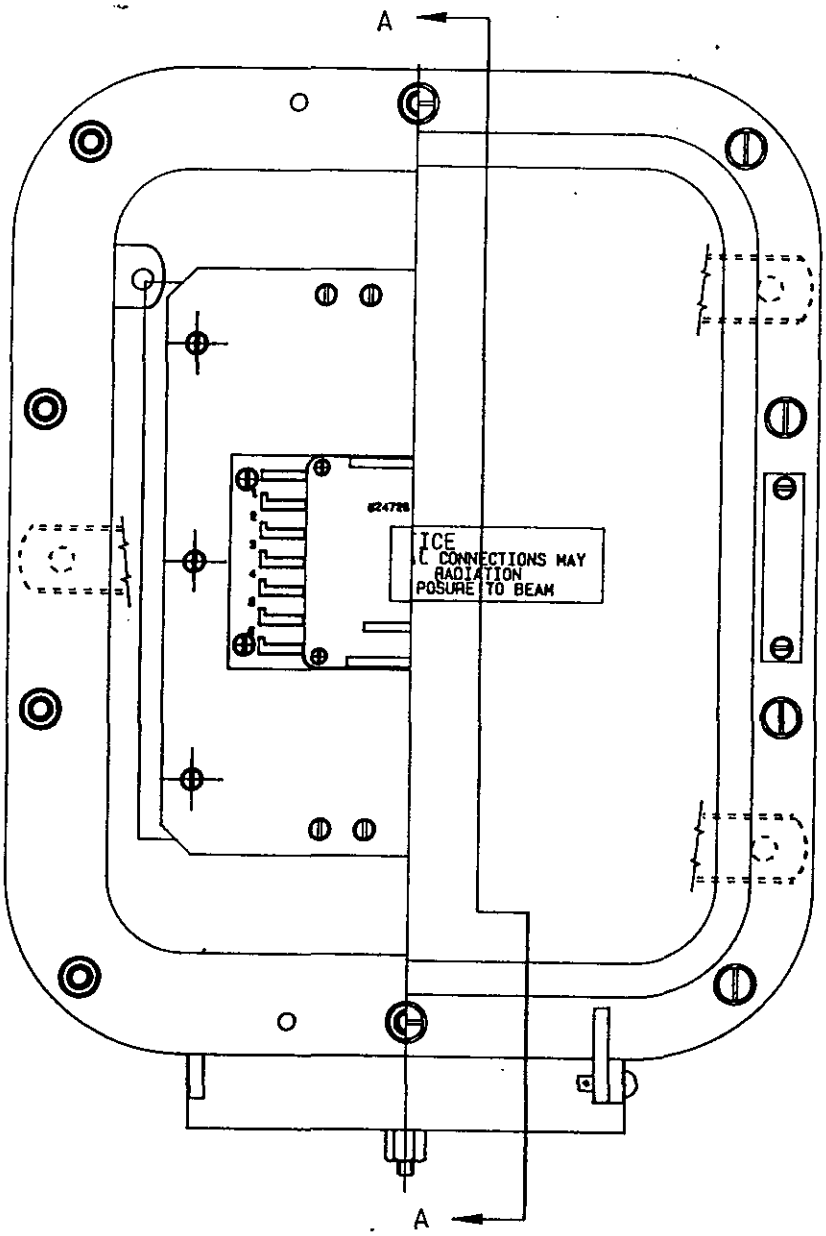


FIGURE 13. Patch panel kit M24728/1-002 installed in an enclosure assembly, M24728/1-001 - Continued.

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ABOVE COVER SHOWN PARTIALLY REMOVED

FIGURE 14. Splice tray kit M24728/1-003 installed in an enclosure assembly, M24728/1-001.

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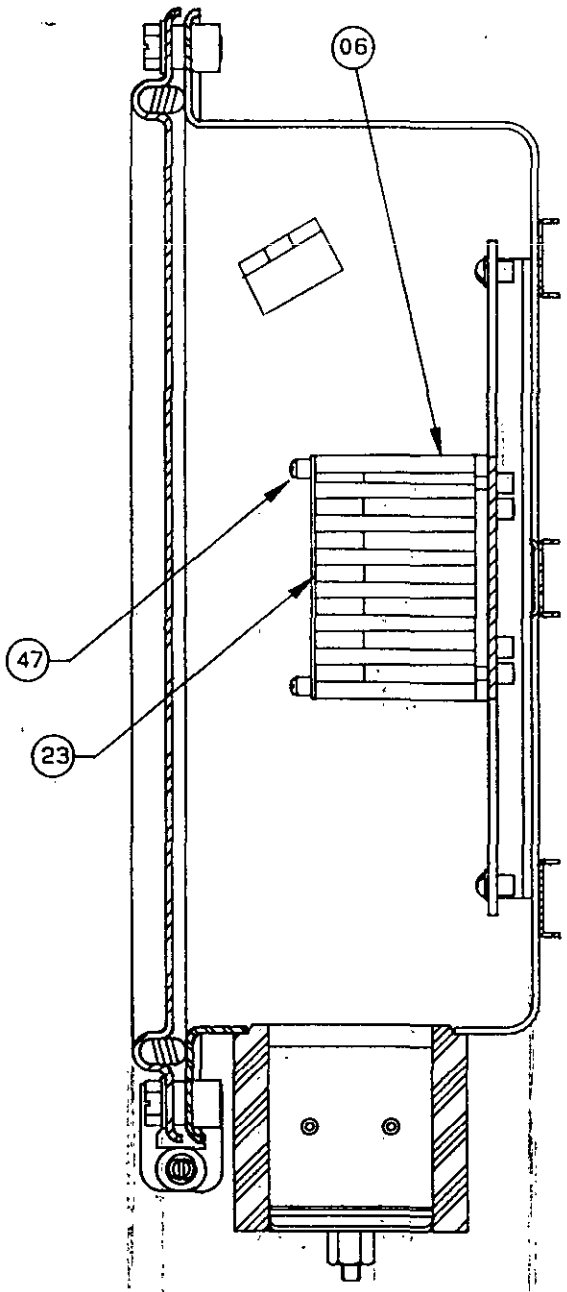
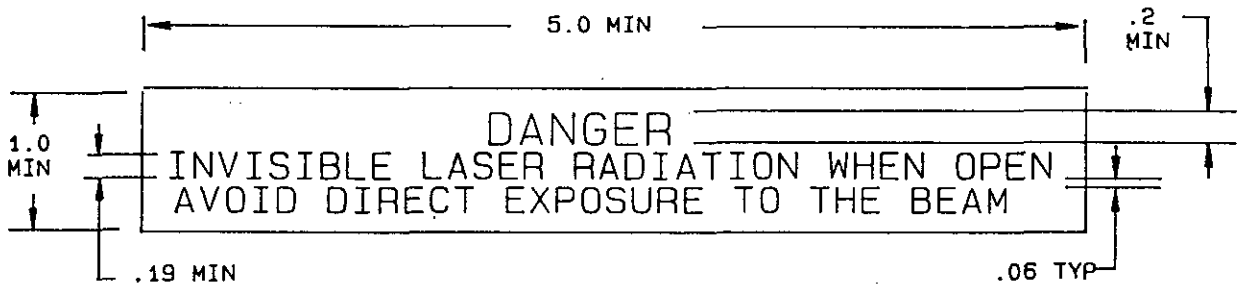


FIGURE 14. Splice tray kit M24728/1-003 installed in an enclosure assembly, M24728/1-001 - Continued.

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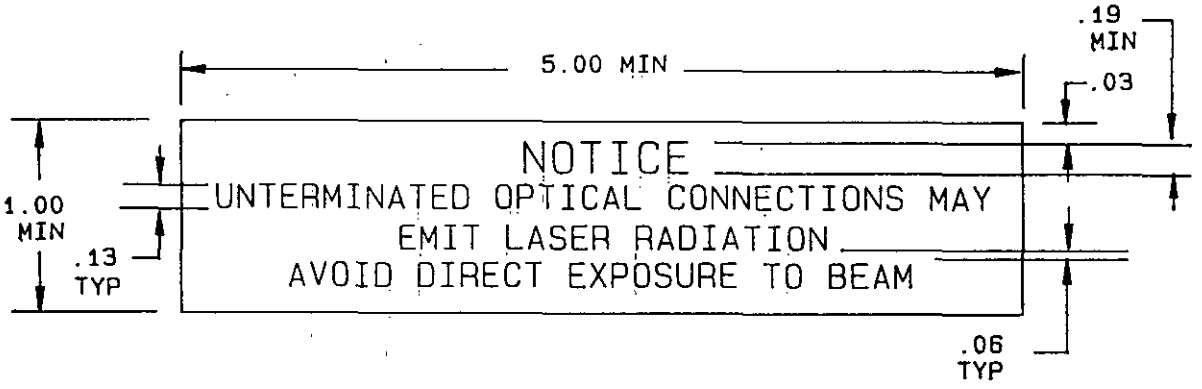


Inches	mm
.06	1.5
.19	4.8
.2	5
1.0	25
5.0	127

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals.

FIGURE 15. Warning label.

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Inches	mm
.03	.76
.06	1.5
.13	3.3
.19	4.8
1.00	25.4
5.00	127.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.01 (0.3) for 2 place decimals.
4. Labels shall be in accordance with MIL-P-15024, type G, adhesive back metal foil, and letters shall be red.

FIGURE 16. Notice label.

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TABLE I. Parts list

Subcomponent identification numbers, circled. (see figures 1 through 16)	Part name	PIN				Subassembly number					Remarks and reference figures
		001	002	003	004	01	02	03	04	05	
	Enclosure assembly	1			1						1
	Patch panel kit		1								13
	Splice tray assembly			1							14
01	Box assembly	1			1	1					1 and 3
02	Cover assembly	1			1		1				1 and 4
03	Mounting plate assembly	1			1			1			1 and 5
04	Slide channel assembly		2						1		8 and 13
05	Patch panel assembly		1							1	10 and 13
06 1/	Splice tray holder			1							6 and 14
07 2/	Wedge pack	1									1
08 3/	6-32 insert									4	6
09 4/	Gasket	1			1						1
10 3/ 4/	Hex bolt	10			10						1
11 5/	Information plate	1			1						1 and 9
12 3/ 6/	Hinge pin	2			2						1
13 3/	1/4 inch flat washer	2			2						1
14 3/	1/16 inch cotter pin	2			2						1
15 3/	3/8 inch flat washer	10			10						1
16 3/	6-32 X 1/4 inch PHMS	4			4						1
17 3/	8-32 X 1/2 inch PHMS	12			12						1
18 3/	# 8 flat washer	12			12						1

See footnotes at end of table.

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TABLE I. Parts list - Continued.

Subcomponent identification numbers, circled. (see figures 1 through 16)	Part name	PIN				Subassembly number					Remarks and reference figures
		001	002	003	004	01	02	03	04	05	
	Enclosure assembly	1			1						1
	Patch panel kit		1								13
	Splice tray assembly			1							14
19 3/	# 8 split lockwasher	12			12						1
20 3/	1/4-20 X 1/2 inch PHMS	8			8						1
21 3/	1/4 inch flat washer	8			8						1
22 3/	1/4 inch split lockwasher	6			6						1
23 1/	Splice tray cover			1							7 and 14
24 7/	MCT frame					1					3
25 5/	10 X 13 box					1					3
26 5/	Hinge					2	2				3 and 4
27 4/ 5/	Flange pad					10					3
28 4/ 5/	Gusset					3					3
29 4/ 5/	Mounting leg					3					3
30 3/	1/4-20 insert					8		2			3 and 5
31 5/	Mounting channel					2					3
32 4/ 5/	10 X 13 cover						1				4
33 4/ 5/	Diagram holder						1				4
34 8/	Mounting plate							1			5
35 3/	8-32 insert							12			5
36 9/	Slide channel								1		8
37 9/	Slide channel bracket								1		8
38 3/	Spring plunger								1		8 and 12
39 8/	Patch panel								1		10
40 3/	1/4-20 captive screw								2		10 and 11
41 3/	3/8-16 helicoil					10					3
42 3/	Door support					1	1				3 and 4
43 10/	Danger label	1			1						1

See footnotes at end of table.

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Check the source to verify that this is the current version before use.

TABLE I. Parts list - Continued.

Subcomponent identification numbers, circled. (see figures 1 through 16)	Part name	PIN				Subassembly number					Remarks and reference figures
		001	002	003	004	01	02	03	04	05	
	Enclosure assembly	1			1						1
	Patch panel kit		1								13
	Splice tray assembly			1							14
44 10/	Notice label	1			1						1 and 16
45 5/	Blank nameplate	1			1						1
46 11/	Cable										1
47 3/	6-32 captive screw			4							7 and 14

Notes (the following notes are for materials specifications only):

1/ Lexan plastic, in accordance with ASTM-D-3935, GE #121, clear or equivalent.

2/ In accordance with MIL-P-24705.

3/ Corrosion resisting steel.

4/ In accordance with MIL-E-24142/6.

5/ Aluminum alloy 356 in accordance with QQ-A-596.

6/ Round head, 1/4 inch dia X 3/4 inch large shank.

7/ Aluminum, 6061-T6 in accordance with ASTM-B-221.

8/ Aluminum, 5052 H-32 in accordance with ASTM-B-209 X .13 thick.

9/ Aluminum, 6061-T6 in accordance with ASTM-B-209 X .13 thick.

10/ In accordance with MIL-P-15024, type G, adhesive-backed, metal foil, red letters.

11/ Aircraft cable, vinyl coated, as required.

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REQUIREMENTS:

Designs and constructions: See figures 1 through 16, and table I.

The interconnection box shall be in accordance with MIL-E-24142/6-004.

Mass: M24728/1-001: 17 max lbs
M24728/1-002: 1 max lbs
M24728/1-003: 1 max lbs
M24728/1-004: 13 max lbs

Interconnection box mounting: See figure 2.

Interconnect organization:

Fiber optic splices: Mechanical/fusion, 72.

Fiber optic adapters: 48 or none.

Fiber optic diagram holders (diagram holder is an integral part of the box): See figure 4.

Cable retention: NA

Cable seal flexing: NA

Cable twist: NA

Compression resistance: NA

Impact: NA

Flammability: NA

Salt spray: NA

Flame spread: NA

Fluid immersion test shall be conducted only on polymeric materials.

Fungus test shall be conducted only on polymeric materials.

The operating temperature range shall be -54 to +65°C and the storage temperature range shall be -62 to +85°C.

Part or Identifying Number (PIN): See table I

M24728/1-001, Box, mounting plate, and lid with cable entrance.

M24728/1-002, Connector panel module.

M24728/1-003, Splice organizer module.

M24728/1-004, Box, mounting plate, and lid without cable entrance.

Qualification by similarity: Manufacturers who produce products for both MIL-I-24728/2 and this specification sheet and are qualified under MIL-I-24728/2, are qualified under this slash sheet.

The gasket size is 10-3/8 wide x 13-3/8 long. This corresponds to PIN M24142/6-004 of MIL-E-24142/6.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Navy - YD, EC, AS
Air Force - 11, 17, 19, 80, 99
DLA - ES

User activities:
Army - AR
Navy - CG, MC

Preparing activity:
Navy - SH

Agent:
DLA - ES

(Project 6099-0001-01)

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